Candidate Seat no………….

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of M. Tech (IT) Examination

December 2015

IT720 Advanced Operating System

Date: 14.12.2015, Monday **Time: 1:30 p.m. To 4:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 **Answer the below given questions:**

a. Give the difference between bounded and unbounded threads. **[02]**

b. Microprocessors were introduced in ______ **[01]**

a) 2nd generation b) 3rd generation c) 4th generation d) 5th generation

c. In Unix, passwords are stored in which of the below given directories? **[01]**

a) bin b) etc c) home d) usr

d. State true or false: 'U area' contains private data manipulated only by kernel. **[01]**

e. To increase the response time and throughput, the kernel minimizes the frequency of disk access by keeping a pool of internal data buffer called **[01]**

a) Pooling b) Spooling c) Buffer cache d) Swapping

f. Special files can be created using system call **[01]**

a) create b) open c) mknod d) both (a) & (c)

Q – 2.a Which are the two modes of Process Execution ? What is a mode switch? **[04]**

Q – 2.b Justify: Operating System is a resource manager. **[05]**

OR

Q – 2.b How is the structure of buffer pool managed? **[05]**

Q – 2.c What are the three ways of I/O Communication? **[05]**

OR

Q – 2.c What is Inode? How does the kernel use the Inode? What are the other data structures used by kernel? **[05]**

Q - 3 **Write short notes:(Any two)** **[14]**

a. Architecture of Unix Systems.

b. Read System Call.

c. Scenarios for retrieval of a buffer.

SECTION II

Q - 4	**Do as Directed:**	
a.	What is a transparent distributed file system?	**[01]**
b.	In most cases, if a process is sent a signal while it is executing a system call : a) the system call will continue execution and the signal will be ignored completely b) the system call is interrupted by the signal, and the signal handler comes in c) the signal has no effect until the system call completes d) None of these	**[01]**
c.	The usual effect of abnormal termination of a program is : a) core dump file generation b) system crash c) program switch d) signal destruction	**[01]**
d.	Which signal is generated when we press control-C? a) SIGINT b) SIGTERM c) SIGKILL d) SIGSEGV	**[01]**
e.	Which of the following signal cannot be handled or ignored? a) SIGINT b) SIGCHLD c) SIGKILL d) SIGALRM	**[01]**
f.	How does a satellite processor handle an open system call?	**[02]**
Q – 5.a	Give the difference between User level threads and Kernel Level threads.	**[04]**
Q – 5.b	What is a signal? What are the default actions taken on signal generation?	**[05]**
	OR	
Q – 5.b	What resources are used when a thread is created in Mach operating system? How do they differ from those when a process is created?	**[05]**
Q – 5.c	What is file locking? Why is it needed? What are the types of file locking?	**[05]**
	OR	
Q – 5.c	Give the difference between named and unnamed pipe. How are they created?	**[05]**
Q – 6	**Write short note:(Any two)**	**[14]**
a.	Semaphore and its requirement in multiprocessor systems.	
b.	Light weight process design issues.	
c.	OPEN System Call.	

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